

CRF: The Walsh–Hadamard Structure of Form Complexity

Four Exact Identities, Parity Balance, and the Spin-0/Spin-1 Decomposition

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Companion to CRF Paper v7 and CRF Clifford series. Discovered through wide pattern mining.

Abstract. The grade complexity vector $\mathbf{g} = (g_0, g_1, g_2, g_3) = (0, 21, 36, 15)$ where $g_k = \binom{d_s}{k}k(n-k)$ encodes the form structure of $\text{Cl}(3)$. Its Walsh–Hadamard transform decomposes into four exact identities:

Symbol	Value	Identity	Physical role
S^+	+72	$n(n+1)$	K35 coupling constant
S^-	−30	$-n_m(n_m+1)$	matter sector charge
S_p	0	0	parity balance (exact, $d_s \geq 2$)
S_3	−42	$-(n-1)(n-2)$	spin-0 vs spin-1 imbalance

Together these four numbers carry *all* information in $\mathbf{g}$, provide an exact reconstruction of the grade contributions, and reveal that spin-1 modes (space+gauge) dominate spin-0 modes (scalar+pseudoscalar) by exactly $(n-1)(n-2) = 42$ in form complexity.

1. The Grade Complexity Vector

The grade complexity $g_k = \binom{d_s}{k}k(n-k)$ for each Clifford grade k :

Grade k	Parity $(-1)^k$	SO(3) spin	Basis	$\binom{3}{k}$	$k(8-k)$	g_k
0	+1	0 (scalar)	$\emptyset$	1	0	0
1	-1	1 (vector)	$\{x\}, \{y\}, \{z\}$	3	7	21
2	+1	1 (axial)	$\{xy\}, \{xz\}, \{yz\}$	3	12	36
3	-1	0 (pseudo)	$\{xyz\}$	1	15	15
Total						72

2. The Walsh–Hadamard Transform

Define four weighted sums of $\mathbf{g}$ using the 4×4 Walsh–Hadamard matrix H with rows $(+1, +1, +1, +1)$, $(+1, +1, -1, -1)$, $(+1, -1, +1, -1)$, $(+1, -1, -1, +1)$:

Four exact identities — complete system

$$S^+ = g_0 + g_1 + g_2 + g_3 = 0 + 21 + 36 + 15 = +72 = n(n+1) \quad (1)$$

$$S^- = g_0 + g_1 - g_2 - g_3 = 0 + 21 - 36 - 15 = -30 = -n_m(n_m+1) \quad (2)$$

$$S_p = g_0 - g_1 + g_2 - g_3 = 0 - 21 + 36 - 15 = 0 \quad (3)$$

$$S_3 = g_0 - g_1 - g_2 + g_3 = 0 - 21 - 36 + 15 = -42 = -(n-1)(n-2) \quad (4)$$

These four values completely determine $\mathbf{g}$ via inverse transform:

$$g_k = \frac{1}{4}(S^+ + H_{1k}S^- + H_{2k}S_p + H_{3k}S_3)$$

Verification: $g_0 = 0$, $g_1 = 21$, $g_2 = 36$, $g_3 = 15$ (exact).

3. Three Proved Identities and One New

3.1. $S^+ = n(n+1)$ — K35 (from CRF Paper v7)

The coupling constant identity: $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$. Proved unique for $(d_s, b, n) = (3, 2, 8)$ by the form consistency theorem.

3.2. $S^- = -n_m(n_m+1)$ — signed K35 (from CRF Clifford)

The grade-sign-weighted sum vanishes on matter sector: $\text{sign}(e_k^2) = +1$ for grades 0,1 and -1 for grades 2,3. Gives matter contribution $= (S^+ - S^-)/2 = 51$ and space

$$= (S^+ + S^-)/2 = 21.$$

3.3. $S_p = 0$ — parity balance (proved, $d_s \geq 2$)

Parity balance — algebraic, holds for all $d_s \geq 2$

$$S_p = \sum_k (-1)^k \binom{d_s}{k} k (b^{d_s} - k) = 0$$

Proof. $S_p = b^{d_s} M_1 - M_2$ where $M_j = \sum_k (-1)^k \binom{d_s}{k} k^j$.

$$M_1 = \frac{d}{dx} (1-x)^{d_s} \Big|_{x=1} = 0 \text{ for } d_s \geq 1.$$

$$M_2 = \frac{d^2}{dx^2} (1-x)^{d_s} \Big|_{x=1} = 0 \text{ for } d_s \geq 2.$$

Therefore $S_p = 0$ for all $d_s \geq 2$ and any integer base b . $\square$

Physical meaning: Even-parity forms (grades 0, 2) and odd-parity forms (grades 1, 3) have equal total form complexity. This is an exact parity symmetry of the differential form algebra, independent of d_s .

3.4. $S_3 = -(n-1)(n-2)$ — spin decomposition (new)

Spin-0 vs spin-1 imbalance — new identity

$S_3 = (g_0 + g_3) - (g_1 + g_2) = (0 + 15) - (21 + 36) = 15 - 57 = -42 = -(n-1)(n-2)$
This measures the excess of spin-1 (vector+axial) over spin-0 (scalar+pseudo) in the form complexity.

Proof of $-(n-1)(n-2)$: $S_3 = g_3 - g_1 - g_2$ (since $g_0 = 0$) $= \binom{3}{3} \cdot 3 \cdot 5 - \binom{3}{1} \cdot 1 \cdot 7 - \binom{3}{2} \cdot 2 \cdot 6 = 15 - 21 - 36 = -42 = -(7)(6) = -(n-1)(n-2)$. $\square$

Physical meaning: Spin-1 modes (space + gauge) carry $57/72 = 79\%$ of total form complexity. Spin-0 modes (scalar + pseudoscalar) carry $15/72 = 21\%$. This spin-1 dominance explains why the CRF crystallisation preferentially excites spin-1 (grade-1 = space) modes: they have more “form weight” and thus more coupling capacity. This is consistent with $C_{\text{mean}} \approx 0.58$ measured in simulation (79% probability in positive-sign sector $\Rightarrow C \approx 2(0.79) - 1 = 0.58$).

4. The Parity–Spin Table

Grade	Parity	Spin	g_k	S^- sign	S_3 sign
0 (scalar)	+1	0	0	+1	+1
1 (vector/space)	-1	1	21	+1	-1
2 (axial/gauge)	+1	1	36	-1	-1
3 (pseudo)	-1	0	15	-1	+1

The S^- signs separate space from matter (grade < 2 vs grade ≥ 2). The S_3 signs separate spin-0 from spin-1. Together they label the four Lorentz multiplets available in 3D: scalar, vector, axial vector, pseudoscalar.

5. Why Space is Grade-1: the Parity–Spin Argument

Among the four grade sectors, only grade-1 satisfies *all* of:

1. Spin-1 (three components, can carry $d_s = 3$ directions)
2. Odd parity (position vector $\mathbf{r} \rightarrow -\mathbf{r}$ under reflection)
3. S^- sign = +1 (positive grade charge, consistent with Phase II)

Grade-0 (spin-0): cannot carry directional information.

Grade-2 (axial): spin-1 but *even* parity; represents rotations, not positions.

Grade-3 (spin-0, center): commutes with all of $\text{Cl}(3)$; cannot distinguish axes.

The $\text{SO}(3)$ attractor in CRF selects *polar vectors* (odd parity, spin-1) as the spatial axes. This is precisely grade-1. The parity and spin together uniquely identify space among the four sectors.

6. Connection to CRF Constants

The interaction range (from K35):

$$\xi = \sqrt{\varepsilon_0} = \frac{\mathcal{G}}{\sqrt{n(n+1)}} = \frac{\mathcal{G}}{\sqrt{S^+}}$$

The coupling constant is the *square root of the total Walsh-Hadamard sum* S^+ .

Summary

The grade complexity vector $\mathbf{g} = (0, 21, 36, 15)$ admits a complete Walsh-Hadamard decomposition into four exact identities:

$S^+ = 72 = n(n+1)$ [K35, specific to $d_s = 3$] $S^- = -30 = -n_m(n_m+1)$ [matter charge] $S_p = 0$ [parity balance, all $d_s \geq 2$] $S_3 = -42 = -(n-1)(n-2)$ [spin-1 dominance]

The parity balance $S_p = 0$ is a universal algebraic fact: even-parity and odd-parity forms always have equal total complexity for $d_s \geq 2$.

The spin imbalance $S_3 = -42$ explains why space (grade-1) dominates CRF dynamics: spin-1 modes carry 79% of form complexity, consistent with simulation ($C_{\text{mean}} \approx 0.58$, $2 \times 0.79 - 1 = 0.58$).

Space is grade-1 because only grade-1 combines spin-1 with odd parity (polar vector), satisfying the $\text{SO}(3)$ requirement for a position field. This bridges the $\text{SO}(3)$ proof with the Clifford grade structure.

The coupling constant is the square root of the Walsh sum: $\mathcal{G} = \sqrt{S^+ \cdot \varepsilon_0}$. The interaction range is the complementary scale: $\xi = \sqrt{\varepsilon_0} = \mathcal{G}/\sqrt{S^+}$.